

VM AI (NEW) - Pretrained Models & Datasets

Complete inventory of models, processors, and datasets used in the VisionMesh AI pipeline

1. Overview

The VM AI (NEW) pipeline evaluates UI designs using a combination of computer-vision and vision-language models. It performs element detection, image captioning, semantic similarity scoring, and layout analysis - all without calling any external API at inference time. Six models are downloaded once and stored locally under VM AI (NEW)/models/ or VM AI (NEW)/CLIP processor/models/.

2. Pretrained Models

#1	BLIP - Bootstrapping Language-Image Pre-training
HF Path:	Salesforce/blip-image-captioning-base
Architecture:	BlipForConditionalGeneration (encoder-decoder ViT + BERT)
Task:	Image Captioning - generates natural-language descriptions of UI screenshots
Datasets:	COCO Captions, Conceptual Captions, SBU Captions, Visual Genome
Loading:	BlipProcessor.from_pretrained() + BlipForConditionalGeneration.from_pretrained() ~900 MB
File Ref:	evaluator/captioner.py : 32-33 download_florence.py : 33
Note: Downloaded via <code>huggingface_hub.snapshot_download()</code> ; stored at <code>models/blip-captioner/</code>	
#2	UIClip - VisionMesh Custom CLIP Fine-tune
HF Path:	keneki21/visionmesh-uiclip (model) keneki21/visionmesh-clip-processor (processor)
Architecture:	CLIPModel (ViT-B/32 backbone, fine-tuned on VisionMesh UI imagery)
Task:	Semantic similarity scoring between UI screenshots and text descriptions
Datasets:	Custom VisionMesh UI dataset (private)
Loading:	CLIPModel.from_pretrained() + CLIPProcessor.from_pretrained() - local with HF Hub fallback
File Ref:	evaluator/scorer.py : 121-134 download_uiclip.py : 15-16
Note: Private HuggingFace repos; requires <code>huggingface_hub.login()</code> before first download	
#3	OpenAI CLIP - Contrastive Language-Image Pre-Training
HF Path:	openai/clip-vit-base-patch32
Architecture:	CLIPModel - Vision Transformer ViT-B/32 + Transformer text encoder
Task:	Base processor/tokeniser backbone for UIClip; general vision-language alignment
Datasets:	YFCC100M (400M image-text pairs from Yahoo Flickr Creative Commons)
Loading:	Config reference; weights stored at CLIP processor/models/clip-processor/
File Ref:	CLIP processor/models/clip-processor/config.json : 2 evaluator/scorer.py
#4	Florence-2 - Microsoft Foundation Vision Model
HF Path:	microsoft/Florence-2-base
Architecture:	Florence2ForConditionalGeneration (DaViT vision encoder + BART-style decoder)

VisionMesh | VM AI (NEW) - Model & Dataset Report

Task:	Element-level captioning and rich spatial understanding of UI components
Datasets:	FLD-5B (Florence Large-scale Detection dataset - 5 billion annotations)
Loading:	<code>AutoProcessor.from_pretrained()</code> + <code>AutoModelForCausalLM.from_pretrained()</code> ~900 MB
File Ref:	<code>models/florence-2-base/config.json</code> <code>download_florence.py</code> <code>main.py</code> : 53

Note: Ships with custom modeling_florence2.py and processing_florence2.py inside the model folder

#5	WebUI ScreenRecognition - UI Element Detector
HF Path:	biglab/webui-screenrecognition-web350k-vins (HuggingFace Space)
Architecture:	TorchScript JIT model (CNN-based object detector, VINS class variant)
Task:	Bounding-box detection of UI elements: buttons, inputs, images, text blocks, etc.
Datasets:	Web350K - 350,000 web UI screenshots with VINS class annotations (16 classes)
Loading:	snapshot_download() then torch.jit.load() ~129 MB (.torchscript file)
File Ref:	evaluator/detector.py : 23 download_models.py : 18-24
#6	CLIP Processor (standalone)
HF Path:	keneki21/visionmesh-clip-processor
Architecture:	CLIPProcessor (BPE tokeniser + ViT-B/32 image pre-processor)
Task:	Tokenisation and image pre-processing for UIClip similarity scoring
Datasets:	Inherits from openai/clip-vit-base-patch32 pre-processing pipeline
Loading:	CLIPProcessor.from_pretrained() - local: CLIP processor/models/clip-processor/
File Ref:	evaluator/scorer.py : 129 CLIP processor/CLIPprocessor.py

3. Datasets Referenced

Dataset	Source	Used For	Description
COCO Captions	Microsoft	BLIP pre-training	~330K images with 5 human captions each. Core captioning benchmark.
YFCC100M	Yahoo / LAION	OpenAI CLIP pre-training	
FLD-5B	Microsoft Research	Florence-2 pre-training	5 billion image-annotation pairs: grounding, OCR, captioning, object detection, and more.
Web350K (VINS)	BigLab / CMU	WebUI ScreenRecognition	
VisionMesh UI Dataset	VisionMesh (private)	UIClip fine-tuning	Custom dataset of UI designs used to fine-tune CLIP for domain-specific UI scoring.
Conceptual Captions	Google	BLIP pre-training	~3.3 million image-caption pairs harvested from web alt-text; complements COCO.
Visual Genome	Stanford	BLIP pre-training	
SBU Captions	Stony Brook U.	BLIP pre-training	~1 million image-caption pairs from Flickr user descriptions.

4. Key Python Dependencies (requirements.txt)

torch >= 2.1.0

Core deep-learning framework; used for all model inference and torch.jit.load()

5. AI Pipeline Flow

Input	UI screenshot (PNG/JPEG) received by Flask endpoint in main.py
Detection	WebUI ScreenRecognition detects bounding boxes of all UI elements -> detector.py
Captioning	BLIP generates a natural-language caption for the full screenshot -> captioner.py
Elem. Caption	Florence-2 captions each detected element region individually
Scoring	UIClip computes cosine similarity between screenshot and caption -> scorer.py

Output	JSON response: element list, captions, similarity scores, composite AI score
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6. Active LLMs - Code Generation Backend (Node.js / Anthropic SDK)

VisionMesh uses two Claude models via the Anthropic API (@anthropic-ai/sdk) for its code generation and report refinement features. These run in the Node.js backend (backend/routes/codeGeneration.js and backend/utis/refineReport.js) and are configurable via environment variables.

#A	Claude Sonnet 4.6 - Code Generation (ACTIVE)
Provider:	Anthropic (cloud API, not self-hosted)
Model ID:	claude-sonnet-4-6
Env Var:	CODEGEN_MODEL (default: claude-sonnet-4-6)
Task:	Generates full multi-file React + Tailwind UI code from a design prompt
SDK:	@anthropic-ai/sdk ^0.98.0 backend/routes/codeGeneration.js
Output:	Structured JSON: 5-file React/Tailwind component scaffold
Note: Primary model for the /generate-code endpoint. Chosen for strong instruction-following and code quality.	

#B	Claude Haiku 4.5 - Report Refinement (ACTIVE)
Provider:	Anthropic (cloud API, not self-hosted)
Model ID:	claude-haiku-4-5-20251001
Env Var:	REFINE_MODEL (default: claude-haiku-4-5-20251001)
Task:	Refines and summarises the raw AI evaluation report into human-readable feedback
SDK:	@anthropic-ai/sdk ^0.98.0 backend/utis/refineReport.js
Output:	Polished markdown report with scores, strengths, and improvement suggestions
Note: Uses Haiku instead of Sonnet for cost efficiency on the lighter refinement task.	

7. LLM Evolution - Models Explored Before Final Stack

Before settling on Anthropic Claude, several LLMs were integrated and evaluated for code generation. The progression below reflects the git commit history of backend/routes/codeGeneration.js (chronological, oldest first).

#E1	Google Gemini 2.0 Flash (explored - removed)
Provider:	Google (cloud API)
Model ID:	gemini-2.0-flash
SDK:	@google/generative-ai ^0.24.1 (still in package.json)
Env Var:	GEMINI_API_KEY, GEMINI_MODEL
Task:	Code generation - React/Tailwind 5-file scaffold from design prompt
Why removed:	Replaced by Groq/free-tier alternatives to avoid API costs

#E2	Groq - Llama 3.3 70B Versatile (explored - removed)
Provider:	Groq (cloud API, free tier)
Model ID:	llama-3.3-70b-versatile
SDK:	groq-sdk ^1.1.2 (still in package.json)
Task:	Code generation with fast inference via Groq's LPU hardware
Why removed:	Hit TPM (tokens-per-minute) rate limits; unreliable for production load

#E3	DeepSeek V3 & DeepSeek R1 (explored - removed)
Provider:	OpenRouter (free tier)
Model IDs:	deepseek/deepseek-chat deepseek/deepseek-r1
SDK:	openai ^6.37.0 with OpenRouter base URL (still in package.json)
Task:	Code generation - evaluated both chat and reasoning variants
Why removed:	Replaced by Together AI and then self-hosted Ollama

#E4	Together AI - Llama 3.3 70B Turbo (explored - removed)
Provider:	Together AI (OpenRouter free tier)
Model ID:	meta-llama/llama-3.3-70b-turbo
SDK:	openai ^6.37.0 with Together AI base URL
Task:	Code generation with fast Llama 3.3 inference
Why removed:	Evaluated alongside Cerebras; both replaced by Ollama for self-hosting

#E5	Cerebras - Llama 3.3 70B (explored - removed)
Provider:	Cerebras (cloud API, free tier)
Model ID:	llama3.3-70b
SDK:	openai ^6.37.0 with Cerebras base URL
Task:	Fast Llama 3.3 70B inference on Cerebras wafer-scale engine
Why removed:	Replaced by self-hosted Ollama approach for offline capability

#E6	Ollama + Qwen2.5-Coder 7B (explored - removed)
Provider:	Self-hosted via Ollama (localhost:11434)
Model ID:	qwen2.5-coder:7b
SDK:	openai ^6.37.0 with Ollama OpenAI-compat. endpoint at /v1
Env Vars:	OLLAMA_URL (default: http://localhost:11434/v1), OLLAMA_MODEL
Task:	Fully offline code generation - no API key or internet required
Why removed:	Replaced by Anthropic Claude for better output quality and reliability
Note: Committed in ccd6d47, deleted in d52c961. OpenAI-compat API layer made swapping easy.	

8. Quick Reference - All Models at a Glance

Vision / Evaluation Models (Python - VM AI (NEW))

Model	HuggingFace ID	Task	Size
BLIP-base	Salesforce/blip-image-captioning-base	Image Captioning	~900 MB
UIClip	keneki21/visionmesh-uiclip	UI Semantic Scoring	~600 MB
CLIP Processor	keneki21/visionmesh-clip-processor	Tokeniser / Pre-proc	~600 MB
CLIP ViT-B/32	openai/clip-vit-base-patch32	Vision-Language Base	~600 MB
Florence-2-base	microsoft/Florence-2-base	Element Captioning	~900 MB
WebUI ScreenRecog.	biglab/webui-screenrecognition-web350k-vins	UI Element Detection	~129 MB

Language Models (Node.js Backend - Code Generation)

Model	Provider / ID	Task	Status
Claude Sonnet 4.6	Anthropic / claude-sonnet-4-6	Code Generation	ACTIVE
Claude Haiku 4.5	Anthropic / claude-haiku-4-5-20251001	Report Refinement	ACTIVE
Gemini 2.0 Flash	Google / gemini-2.0-flash	Code Generation	Explored
Llama 3.3 70B	Groq / llama-3.3-70b-versatile	Code Generation	Explored
DeepSeek V3	OpenRouter / deepseek/deepseek-chat	Code Generation	Explored
DeepSeek R1	OpenRouter / deepseek/deepseek-r1	Code Generation	Explored
Llama 3.3 70B Turbo	Together AI / Llama-3.3-70B-Turbo	Code Generation	Explored
Llama 3.3 70B	Cerebras / llama3.3-70b	Code Generation	Explored
Qwen2.5-Coder 7B	Ollama / qwen2.5-coder:7b (self-hosted)	Code Generation	Explored

Vision models run fully offline (Python/Flask). Language models call the Anthropic cloud API. All explored LLMs are removed from active code but preserved in git history.